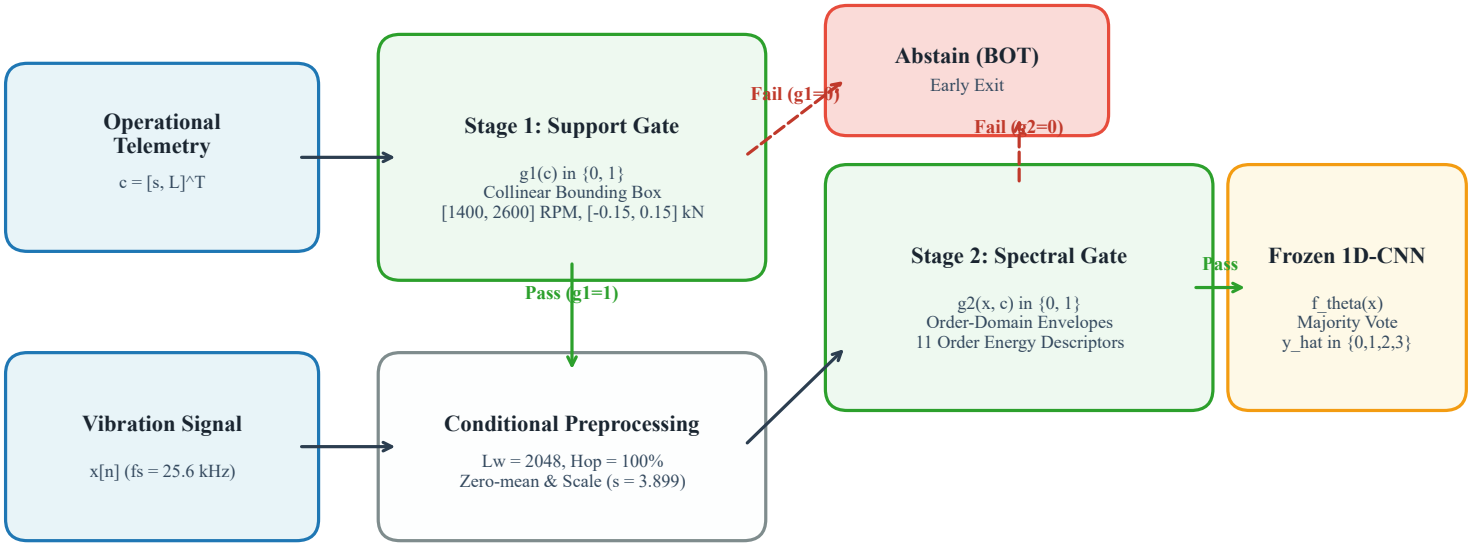

Operational Telemetry
c = [s, L]^T
Stage 1: Support Gate
g1(c) in {0, 1}
Collinear Bounding Box
[1400, 2600] RPM, [-0.15, 0.15] kN
Fail (g1=0)
Abstain (BOT)
Early Exit
Pass (g1=1)
Fail (g2=0)
Vibration Signal
x[n] (fs = 25.6 kHz)
Conditional Preprocessing
Lw = 2048, Hop = 100%
Zero-mean & Scale (s = 3.899)
Stage 2: Spectral Gate
g2(x, c) in {0, 1}
Order-Domain Envelopes
11 Order Energy Descriptors
Pass
Frozen 1D-CNN
f_theta(x)
Majority Vote
y_hat in {0,1,2,3}